

Skills Summary

Converting Between Rectangular and Polar Coordinates

Use this reference while completing your worksheet or when studying.

Skill 1 — Polar to rectangular

$$x = r \cos \theta, \quad y = r \sin \theta.$$

Read them off the right triangle with legs x and y and hypotenuse r : the leg beside θ is $r \cos \theta$, the leg opposite it is $r \sin \theta$.

This direction is always safe — no quadrant decision is needed, because the signs of $\cos \theta$ and $\sin \theta$ already carry it.

Example: Convert the polar point $(10, 30^\circ)$ to rectangular form.

Step 1. Going from polar to rectangular means substituting into the two formulas — nothing to decide, so this direction is the easy one.

$$\text{Step 2. } x = 10 \cos 30^\circ = 10 \left(\frac{\sqrt{3}}{2} \right), \quad y = 10 \sin 30^\circ = 10 \left(\frac{1}{2} \right) \Rightarrow (x, y) = (5\sqrt{3}, 5)$$

Try it: Convert $(12, 150^\circ)$ to rectangular form.

$$y = 6\sqrt{3}, x = -6$$

Skill 2 — Rectangular to polar

$$r = \sqrt{x^2 + y^2} \quad (\text{always taken positive here}), \quad \tan \theta = \frac{y}{x}.$$

The r formula is the Pythagorean theorem on the same triangle. Sketch the point first: knowing its quadrant before you touch the calculator is what makes the angle step reliable.

Example: Convert the rectangular point $(4, 4\sqrt{3})$ to polar form.

Step 1. Both coordinates are positive, so the point is in quadrant I and the calculator's inverse tangent will need no correction.

$$\text{Step 2. } r = \sqrt{16 + 48} = \sqrt{64}, \quad \tan \theta = \frac{4\sqrt{3}}{4} = \sqrt{3} \Rightarrow r = 8, \quad \theta = 60^\circ$$

Try it: Convert $(5, 5)$ to polar form with $r > 0$.

$$\text{check: } r = 5\sqrt{2}, \theta = 45^\circ$$

Skill 3 — Choosing the angle by quadrant

$\tan^{-1}$ only ever returns an angle between -90° and 90° , so it can only land in quadrant I or IV. Tangent repeats every 180° , so opposite directions share the same ratio.

Rule: if $x < 0$, add 180° to the calculator's answer. If $x > 0$ and $y < 0$, add 360° to bring it into $[0^\circ, 360^\circ)$. If $x = 0$ the formula does not apply — read the angle off the axis.

Example: Convert $(-3, -3\sqrt{3})$ to polar form with $0^\circ \leq \theta < 360^\circ$.

Step 1. Both coordinates are negative, so the point is in quadrant III, and the inverse tangent cannot report an angle there — expect to correct it.

Step 2. $r = \sqrt{9 + 27} = 6$; $\tan^{-1}\left(\frac{-3\sqrt{3}}{-3}\right) = 60^\circ$, and $x < 0$, so add 180° . $\Rightarrow r = 6$, $\theta = 240^\circ$

Try it: Convert $(-4, 4)$ to polar form with $0^\circ \leq \theta < 360^\circ$.

check: $r = 4\sqrt{2}$, $\theta = 135^\circ$

Skill 4 — One point, many names

A polar name is not unique. The same point is (r, θ) , $(r, \theta + 2\pi)$, and also $(-r, \theta + \pi)$.

Why the last one works: a negative first coordinate means face the given direction and step *backwards*, which lands you where facing the opposite direction and stepping forwards would.

Example: Rename the polar point $\left(4, \frac{\pi}{6}\right)$ so that its first coordinate is -4 .

Step 1. Flipping the sign of the distance reverses the direction, so the angle must turn by a straight angle to compensate.

Step 2. $\theta' = \frac{\pi}{6} + \pi = \frac{\pi}{6} + \frac{6\pi}{6} \Rightarrow \theta' = \frac{7\pi}{6}$, so the point is also $\left(-4, \frac{7\pi}{6}\right)$.

Try it: Rename $\left(3, \frac{3\pi}{4}\right)$ so that its first coordinate is -3 .

check: $\theta = \frac{7\pi}{4}$